

NutNet Multidimensional Stability Analyses

This repository contains the R code and analytical workflows accompanying the manuscript:

Yang et al. Short-term gains mask long-term instabilities from grassland enrichment.

The repository is organised as a series of self-contained R Markdown (.Rmd) workflows that reproduce the data preparation, stability calculations, statistical analyses, and figures presented in the manuscript.

R Markdown workflows

- **01_prepare_analytical_dataset.Rmd**

Processes the raw Nutrient Network (NutNet) biomass, cover, and soil datasets, applies the inclusion criteria described in the manuscript, harmonises plant taxonomy within sites, and constructs the cleaned analytical dataset (df.tidy) used throughout subsequent analyses.

This document is not required to reproduce the analyses and figures reported in the manuscript because the processed analytical datasets are already provided in this repository. Readers who wish to reproduce the complete data-cleaning workflow should first obtain the corresponding raw NutNet datasets from the Nutrient Network consortium. Running the remaining workflows is not affected by omitting this step.

- **02_calculate_stability_metrics.Rmd**

Calculates the four stability components analysed in the manuscript: functional temporal variability, extent of functional change, compositional temporal variability, and extent of compositional change. This workflow also reproduces the distribution and correlation figures of the stability metrics.

- **03_analyse_single_stability_responses.Rmd**

Evaluates how nutrient enrichment intensity influences each individual stability component using linear mixed-effects models.

- **04a_check_temporal_effects_on_dimensionality.Rmd**

Performs preliminary permutation analyses to evaluate whether temporal window length affects the dimensionality of ecological stability and whether temporal effects interact with enrichment intensity.

- **04b_enrichment_effects_on_stability_dimensionality.Rmd**

Performs the main dimensionality analyses, including generalized least-squares models and permutation tests, to evaluate how nutrient enrichment intensity influences the multidimensional organisation of ecological stability.

- **05_reproduce_remaining_figures_and_sensitivity_analyses.Rmd**

Reproduces the remaining supplementary figures and sensitivity analyses, including legume relative abundance, species richness responses, dominance responses, baseline soil N:P ratio analyses, directional biomass responses, site-level maps of nutrient enrichment intensity, and sensitivity analyses of single stability responses.

Data availability

This repository contains the curated datasets necessary to reproduce the analyses presented in the manuscript. The complete raw Nutrient Network database is maintained by the Nutrient Network consortium and is available upon request according to NutNet data-sharing policies.

The analytical datasets included here represent the processed data used in the manuscript and are sufficient to reproduce all reported analyses and figures.

Software requirements

All analyses were conducted in R version 4.4.2 on a Windows 10 (64-bit) workstation with 16 GB RAM. The versions of all analytical R packages used in this study are provided here. To render the R Markdown documents to PDF, users additionally require the packages rmarkdown, knitr, and a LaTeX installation such as TinyTeX.

Package	Version	Package	Version
tidyverse	2.0.0	dlookr	0.6.3
magrittr	2.0.3	GGally	2.2.1
grid	4.4.2	PerformanceAnalytics	2.0.4
gridExtra	2.3	glmmTMB	1.1.10
rnaturalearth	1.0.1	ggeffects	1.7.2
rnaturalearthdata	1.0.0	patchwork	1.3.2
ggcorrplot	0.1.4.1	RColorBrewer	1.1.3
scales	1.4.0	lme4	1.1.35.5
sp	2.1.4	lmerTest	3.1.3
sf	1.0.19	cowplot	1.1.3
broom	1.0.7	gtable	0.3.6
broom.mixed	0.2.9.6	rgbif	3.8.4
nlme	3.1.166	sjPlot	2.8.16
scico	1.5.0	pacman	0.5.1
vegan	2.6.10		

Installation

No software installation beyond a standard R environment is required. Users should place the repository contents within a single working directory and preserve the folder structure used in the repository.

Instructions for use

The analyses are intended to be executed sequentially:

- 01_prepare_analytical_dataset.Rmd
- 02_calculate_stability_metrics.Rmd
- 03_analyse_single_stability_responses.Rmd
- 04a_check_temporal_effects_on_dimensionality.Rmd
- 04b_enrichment_effects_on_stability_dimensionality.Rmd
- 05_reproduce_remaining_figures_and_sensitivity_analyses.Rmd

Each R Markdown workflow contains (i) an Overview section describing the scientific objective, (ii) an Output section describing the generated figures and datasets, (iii) detailed annotations explaining each analytical step, and (iv) self-contained code chunks that minimise dependencies among sections. The knitted PDF files provide a human-readable record of all analyses and outputs.

Demonstration examples

Because the repository is organised as a collection of reproducible R Markdown workflows rather than a software package, separate demonstration datasets are not provided. The workflows themselves serve as executable demonstrations of the analyses.

The dimensionality analyses in 04a_check_temporal_effects_on_dimensionality.Rmd and 04b_enrichment_effects_on_stability_dimensionality.Rmd use permutation procedures that can be computationally intensive (approximately 15 minutes per workflow on the machine described above). For demonstration purposes, users may reduce the number of permutation iterations (for example, from 1,000 to 100) to substantially shorten computation time. Reducing the number of permutations may slightly decrease the precision of the estimated null distributions and associated significance tests.

Reproducibility statement

This repository was developed to satisfy the reproducibility requirements of the Nature Portfolio Software and Code Submission Checklist. Specifically, the repository provides: (i) the code required to reproduce the manuscript analyses and figures; (ii) documentation of the analytical workflow and software requirements; (iii) instructions for use; (iv) executable examples embedded within the R Markdown workflows; (v) guidance for reduced-run demonstrations of computationally intensive analyses; and (vi) extensive annotations to facilitate understanding, verification, and reuse.